

OPT-6.7B — Hallucination Detection Comparison

Each cell shows ROC-AUC. Per column: **1st place (red)**, **2nd (green)**, **3rd (blue)**.

Wikipedia = held-out in-distribution (n=400); downstream datasets evaluated via multi-task pipeline (SAPLMA/HaloScope/HalluShift full eval; EigenScore subsampled to 100 records/dataset).

Method	Feature stack / source	Wikipedia	TruthQA	TrivQA	CoQA	TydiQA	HE-QA	HE-Sum	HE-Dial	avg
Variant A	canonical (MIND)	0.489	0.328	0.535	0.571	0.503	0.054	0.423	0.345	0.406
Variant B	+ D_{mean}	0.494	0.338	0.567	0.564	0.523	0.039	0.316	0.313	0.394
Variant C	+ V_{last}	0.485	0.308	0.561	0.550	0.537	0.025	0.321	0.310	0.387
Variant D	+ H_{mean}	0.515	0.303	0.539	0.556	0.542	0.126	0.342	0.336	0.407
Variant E	+ D_{mean} + V_{last} + H_{mean} (MIND+)	0.461	0.276	0.485	0.502	0.492	0.228	0.380	0.342	0.396
Variant F	+ F1 (Lookback Ratio)	0.488	0.286	0.546	0.558	0.506	0.032	0.361	0.316	0.387
Variant G	+ F5 (Logit-Lens JSD)	0.496	0.313	0.527	0.542	0.514	0.026	0.336	0.309	0.383
Variant H	+ F7 (Max-Margin)	0.494	0.308	0.528	0.559	0.516	0.037	0.335	0.313	0.386
Variant I	+ F1 + F5 + F7 (trio)	0.538	0.303	0.519	0.551	0.504	0.055	0.334	0.349	0.394
Variant J	+ all 9 F-features	0.427	0.308	0.626	0.562	0.567	0.246	0.395	0.353	0.436
Variant K	MIND+ + trio	0.472	0.328	0.553	0.551	0.538	0.049	0.286	0.302	0.385
Variant L	MIND+ + all 9 F (everything)	0.455	0.293	0.528	0.554	0.498	0.020	0.352	0.318	0.377
SAPLMA	MLP probe on layer-20 hidden (Azaria & Mitchell 2023)	0.371	0.529	0.419	0.468	0.517	0.795	0.490	0.429	0.502
HaloScope	Spectral + non-linear probe (Du et al. NeurIPS 2024)	0.458	0.665	0.551	0.435	0.420	0.291	0.264	0.382	0.433
HalluShift	Wasserstein/cosine + token-prob (Dasgupta IJCNN 2025)	0.512	0.660	0.602	0.543	0.526	0.456	0.536	0.450	0.536
EigenScore	K-sample log-det covariance (Chen et al. INSIDE, ICLR 2024)	0.465	0.490	0.584	0.458	0.580	0.500	0.499	0.499	0.509

Reading the table. Higher AUROC = better detector. Rows 1–12 are the 12 ablation variants on top of the canonical MIND backbone (last-token / last-layer hidden state, 4096-dim for OPT-6.7B). Rows 13–16 are paper-faithful re-implementations of four SOTA hallucination detectors, all evaluated on the SAME 80/20 split of the SAME MIND-generated dataset (1000 hallucinated + 1000 non-hallucinated, seed=42).

Variant key. A = canonical (no addition). B/C/D = adds D_{mean} (layer-drift) / V_{last} (cross-layer variance) / H_{mean} (predictive entropy) individually. E = adds all three (MIND+ headline). F/G/H = adds F1 (Lookback Ratio) / F5 (Logit-Lens JSD) / F7 (Max-Margin) individually. I = the F1+F5+F7 trio. J = all 9 F-features. K = MIND+ stack plus the trio. L = MIND+ stack plus all 9 F-features.

Headline. **HalluShift** wins on average across the 8 columns (avg AUROC = 0.536), followed by EigenScore (0.509) and SAPLMA (0.502). The strongest *variant* is **J** (avg = 0.436), which uses all 9 novel F-features but does not include the MIND+ 3-feature triplet; it wins TriviaQA (0.626) outright and places 2nd on TydiQA. None of the supervised variants beats HalluShift on the average, but on individual datasets specific variants surface as top-3.